

The RAG Categories Cheat Sheet

Six flavors of RAG, and when each one earns its place in a pipeline

Several learners asked for a single tabular view of the different RAG categories to screenshot for later. This card is that reference: the six categories covered in Session 4, what each one actually does, and where it fits in a real pipeline. Keep the Session Notes for the full explanations and examples behind each row.

Category	How it works	Best for	Key tools
Vector RAG (naive)	Embeds chunks and retrieves the top K by cosine similarity	Semantic, meaning based questions	Qdrant, ChromaDB, Milvus
Keyword RAG (vector less)	Matches on keywords using BM25 or TF IDF, with no embeddings at all	Code, jargon, IDs, citations, and precise keyword queries	Elasticsearch, Apache Solr, Apache Lucene
Hybrid RAG	Runs vector search and keyword search in parallel and merges the results	Most production systems today	Elasticsearch or Solr plus any vector database
Graph RAG	Builds a knowledge graph of entities and relationships, then traverses it	Relationship style questions, for example which companies a person founded	Graph databases, knowledge graph tooling
SQL RAG (Text to SQL)	An LLM writes a SQL query, the query runs against a real database, and the result is explained back in plain language	Structured business data already sitting in a SQL database	Any relational database plus an LLM
Reasoning based RAG (PageIndex)	Builds a hierarchical index of a long document's sections, then traverses just that index	Very long documents, still experimental	PageIndex, an early stage open source library

Key Insight

Hybrid RAG and SQL RAG together are estimated to cover 80 to 90 percent of real production use cases, which is exactly why they are the two categories getting hands on treatment next week. Graph RAG and reasoning based RAG solve narrower problems and are used far less often in practice.

How to Use This Card

Start with the question you are trying to answer, not the tool you already know. If the answer depends on meaning, start with vector RAG. If it depends on an exact code, ID, or citation, add keyword search. If your data already lives in a SQL database, reach for SQL RAG before building any retrieval pipeline at all.